github.com/jeronimoluza
avraltod@gmail.com

–

Δ

o_H, o_D, o_A

$$p_r = \frac{1/o_r}{\sum_{s\in\{H,D,A\}} 1/o_s}, r \in \{H, D, A\},$$

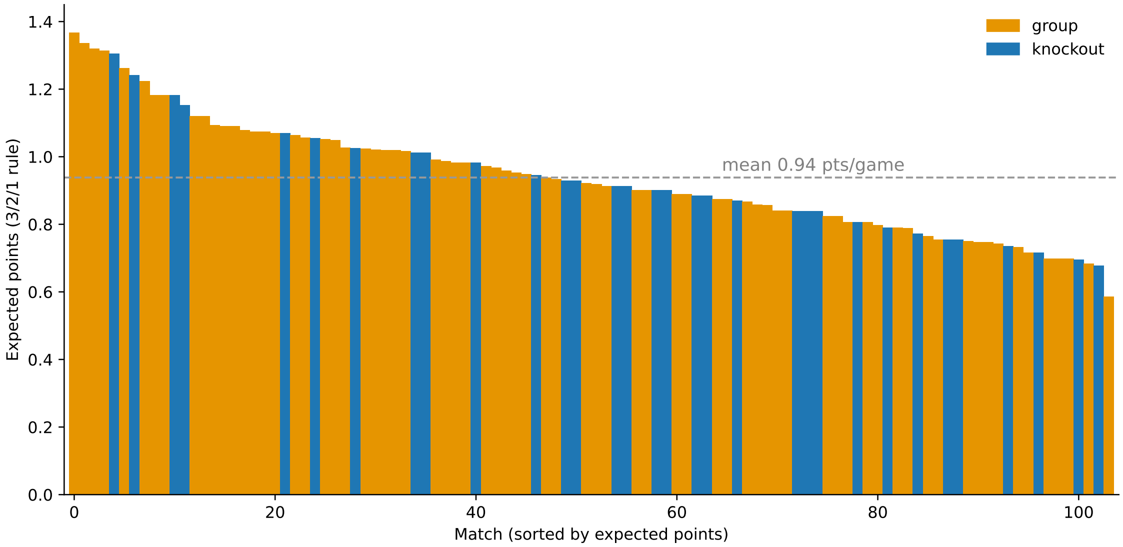
λ_h, λ_a

$$(\hat{\lambda}_h, \hat{\lambda}_a) \ = \ _{\lambda_h,\lambda_a} \sum_{r\in\{H,D,A\}} \big(\pi_r(\lambda_h, \lambda_a) - p_r\big)^2 1.6 \le \lambda_h + \lambda_a \le 3.4,$$

$\pi_r(h, a) r g$

$$(h, a) \ = \ P(h, a) \ + \ P\big(= r, \ = g\big) \ + \ P\big(= r\big),$$

$(\hat{\lambda}_h, \hat{\lambda}_a)$



$$E_{AB} = \left(1 + 10^{-\Delta/400}\right)^{-1}, \Delta = R_A - R_B,$$

$p_D = 0.30\, e^{-|\Delta|/700} p_H = E_{AB} - p_D/2 R{-}20{-}15{-}10{-}10{-}5{+}40{+}20$
$N = 200{,}000 p \sqrt{p(1-p)/N} \leq 0.11$

$-N$

s

$$T_s^* \;=\; {}_{T}\; P\big(T s\big),$$

qp

$$D(q \,\|\, p) \;=\; \sum_t q_t \, {}_2 \, \frac{q_t}{p_t}$$

D

$D(q \,\|\, p) q$

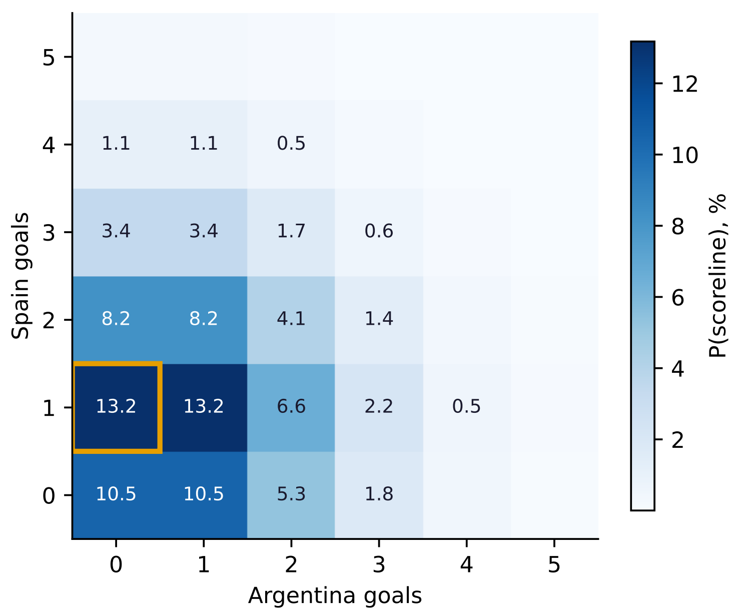
$\rightarrow$

$p_h + p_d/2 0.30$

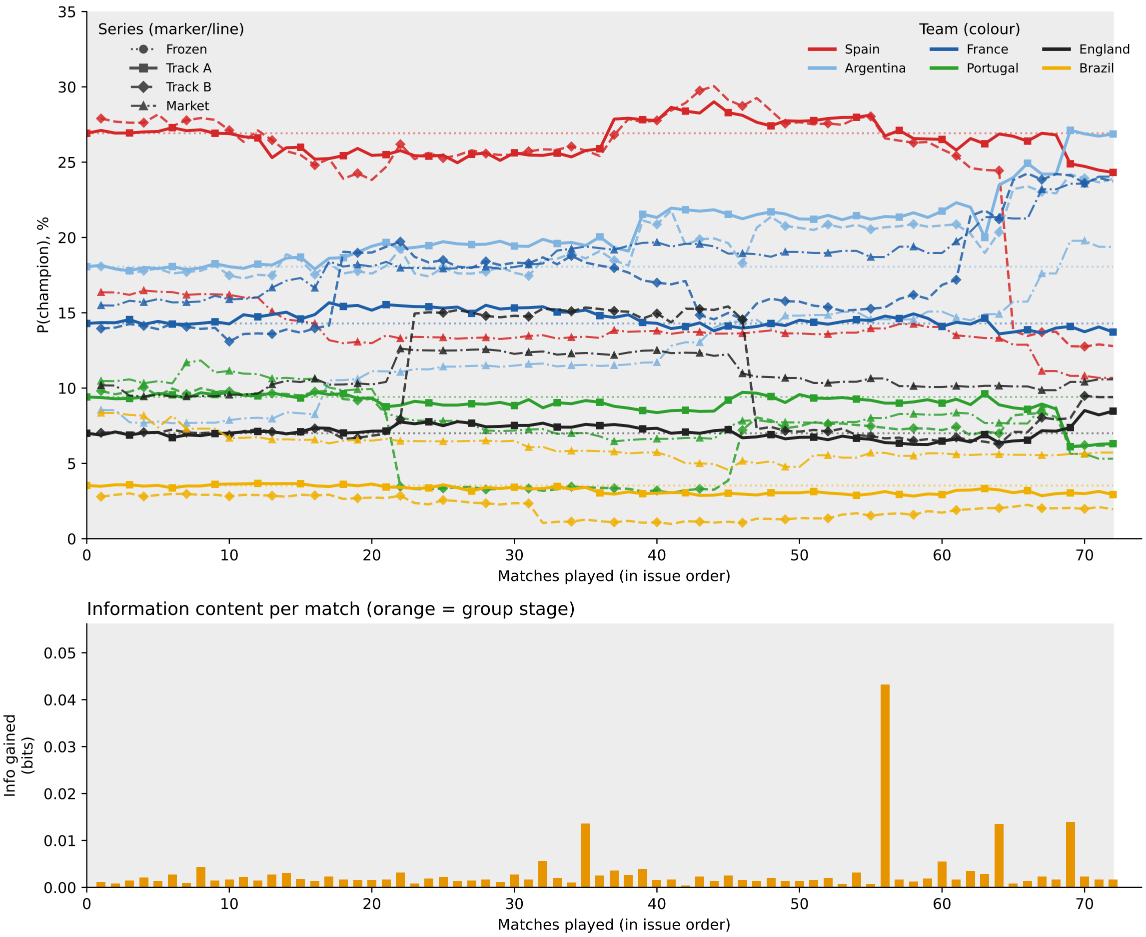
$\Delta -$

N`data/frozen_stage_probs.json`$N\Delta -$

$\rightarrow$ $+4.0$
$\rightarrow$ $+2.4$
$\rightarrow$ $+9.6$



$\hat{\lambda} = 1.25 \hat{\lambda} = 1.00$

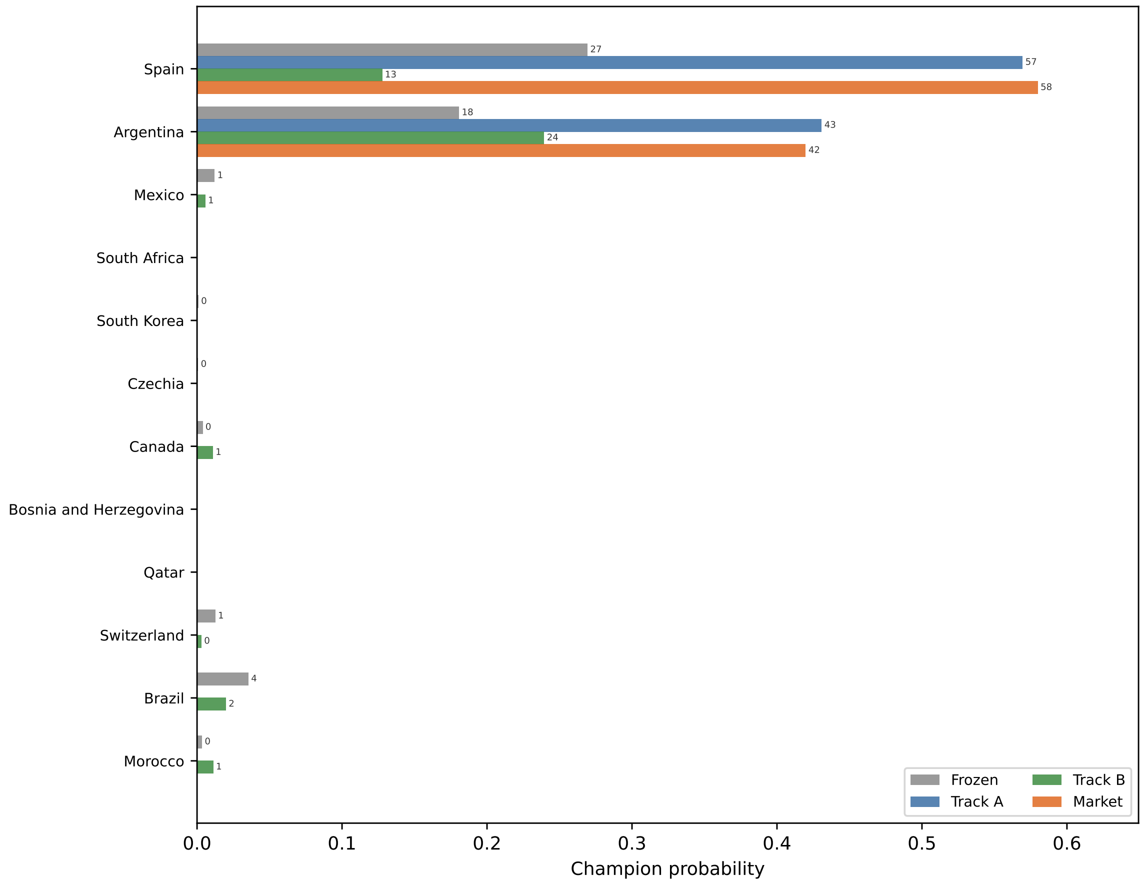


Series (marker/line)
Frozen
Track A
Track B
Market
Team (colour)
Spain
Argentina
France
Portugal
England
Brazil
P(champion), %
Matches played (in issue order)
Information content per match (orange = group stage)
Info gained (bits)
Matches played (in issue order)

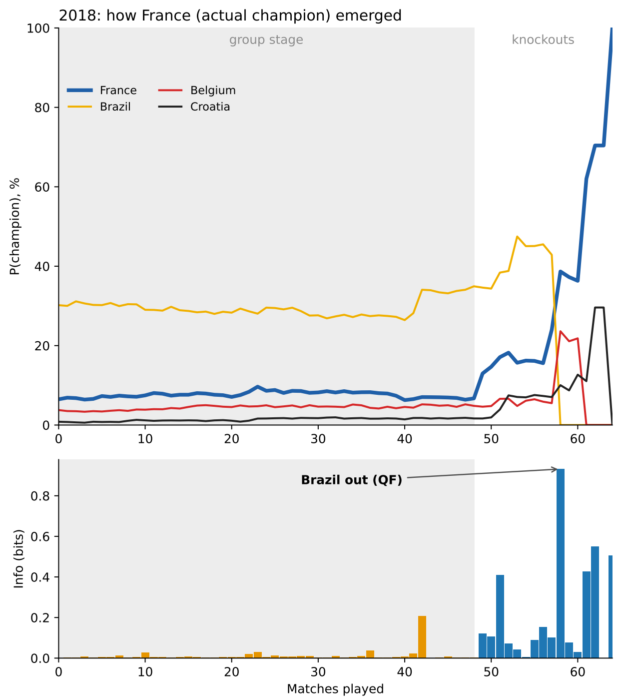
Δ−	Δ−	
−14.2	+1.1	−75.5
+5.9	+1.1	−58.6
+0.1	+42.6	+92.9
+0.3	+54.7	+40.5
−0.2	+23.9	−56.8
+0.5	+3.6	+56.1
	+37.6	−0.1
−1.5	+1.6	−48.7
+0.7	+11.6	+90.4
	+65.3	+24.1
+4.0	+6.0	+126.0
	+71.0	+57.8
−1.5	+4.6	−99.6
	−7.0	−55.2
	−68.3	−86.2
+0.3	+12.3	+23.6
	+23.4	+8.8
+2.5	+1.0	−12.8
+9.3	+1.5	+50.1
+1.8	+1.2	+54.5
	+77.1	+107.1
	−12.0	+33.5
−0.1	−62.6	−52.6
	−7.7	−48.5
	+18.6	−22.9
−0.6	+24.0	−64.4
	−28.5	+22.4
−0.6	+8.4	−13.5
+0.8	+11.2	+79.3
−1.8	+9.0	−68.5
	−30.9	−12.2
+0.3	+10.6	−17.6
	−35.9	+5.3
−0.3	+34.8	−115.4
−3.2	+2.7	−96.2
	−24.3	+2.9
	−34.0	+17.0
	−69.1	−64.1
	+18.6	+16.0

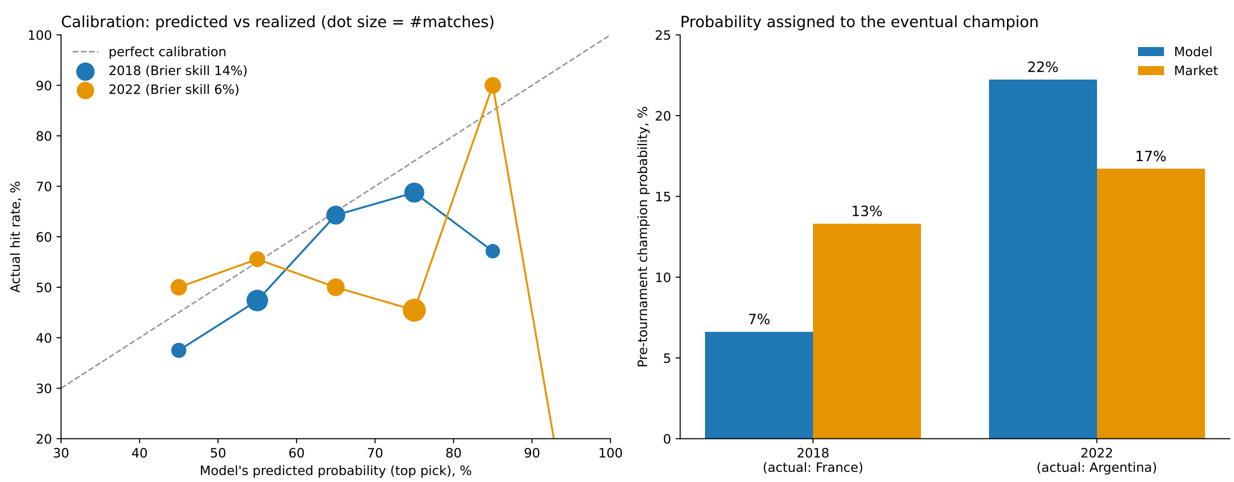
$\Delta-$		$\Delta-$	
		$+54.7$	$+154.8$
		-66.2	-55.1
$+0.8$		$+35.8$	$+161.0$
-1.0		$+6.0$	-93.1
		-40.2	-28.2
		-80.4	-51.2
$+0.3$		$+17.7$	$+126.1$
		-87.8	$+34.4$
		-38.5	-87.6

$> 0.50 < 25$
$\lambda = 1.20 \lambda = 1.05$
$\lambda = 1.75 \lambda = 0.60$
$\lambda = 0.85 \lambda = 1.60$
$\lambda = 1.65 \lambda = 0.80$
$\lambda = 0.10 \lambda = 2.00$
$\lambda = 0.45 \lambda = 1.85$
$\lambda = 1.25 \lambda = 1.00$
$\lambda = 0.20 \lambda = 1.80$
$\lambda = 2.20 \lambda = 0.10$
$\lambda = 1.75 \lambda = 0.70$
$\lambda = 1.70 \lambda = 0.75$
$\lambda = 1.85 \lambda = 0.10$
$\lambda = 1.60 \lambda = 0.80$
$\lambda = 1.70 \lambda = 0.70$
$\lambda = 2.05 \lambda = 0.10$
$\lambda = 1.85 \lambda = 0.20$
$\lambda = 1.60 \lambda = 0.85$



$\lambda = 1.05 \lambda = 1.15$
$\lambda = 2.15 \lambda = 0.10$
$\lambda = 0.80 \lambda = 1.60$
$\lambda = 1.00 \lambda = 1.40$
$\lambda = 1.20 \lambda = 1.05$
$\lambda = 1.65 \lambda = 0.80$
$\lambda = 1.05 \lambda = 1.05$
$\lambda = 1.45 \lambda = 0.95$
$\lambda = 1.60 \lambda = 0.85$
$\lambda = 1.05 \lambda = 1.05$
$\lambda = 1.00 \lambda = 1.35$
$\lambda = 1.00 \lambda = 1.25$
$p_o q_o \sum_o p_o \, D(q_o \parallel q) q q = \sum_o p_o \, q_o$



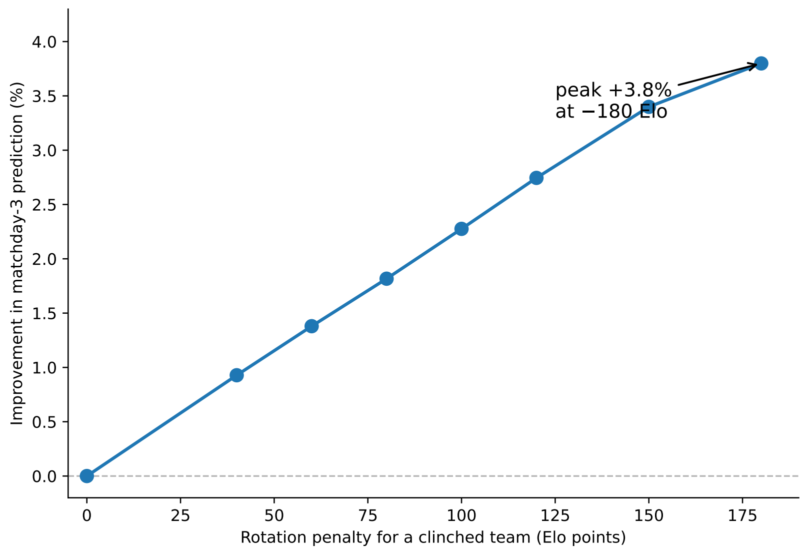


-5.3

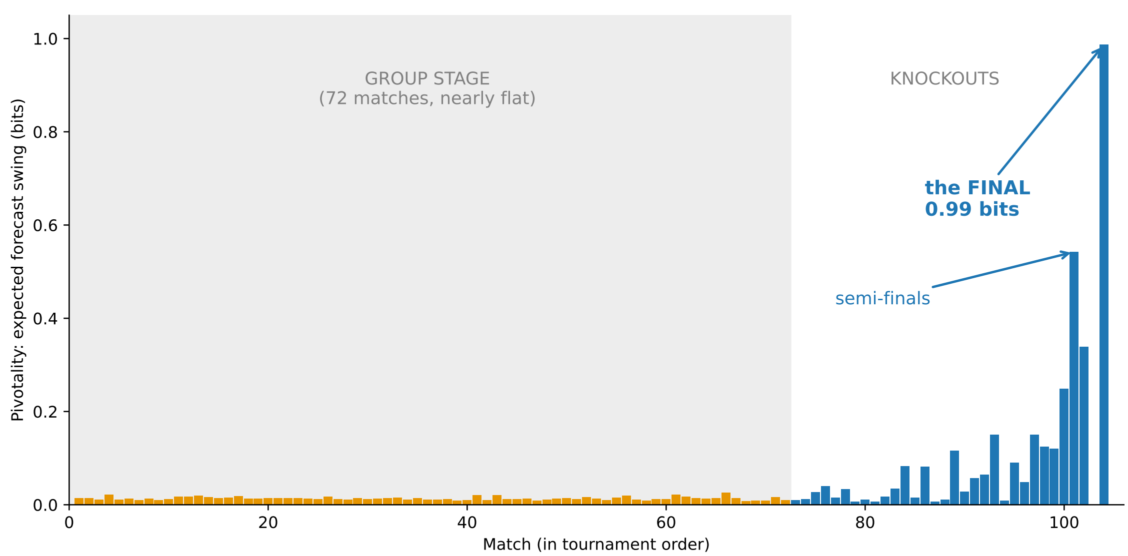
-0.09

λ

$\rightarrow$

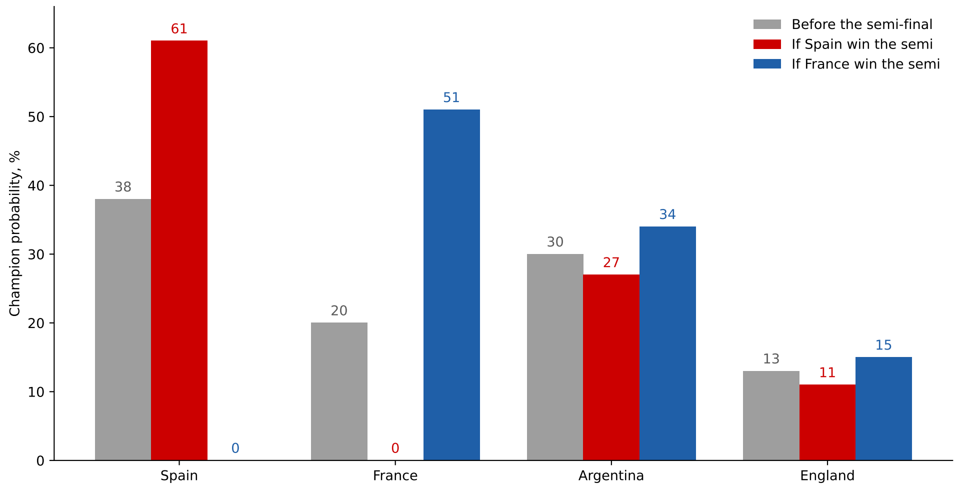


peak +3.8%
at −180 Elo
Improvement in matchday-3 prediction (%)
Rotation penalty for a clinched team (Elo points)

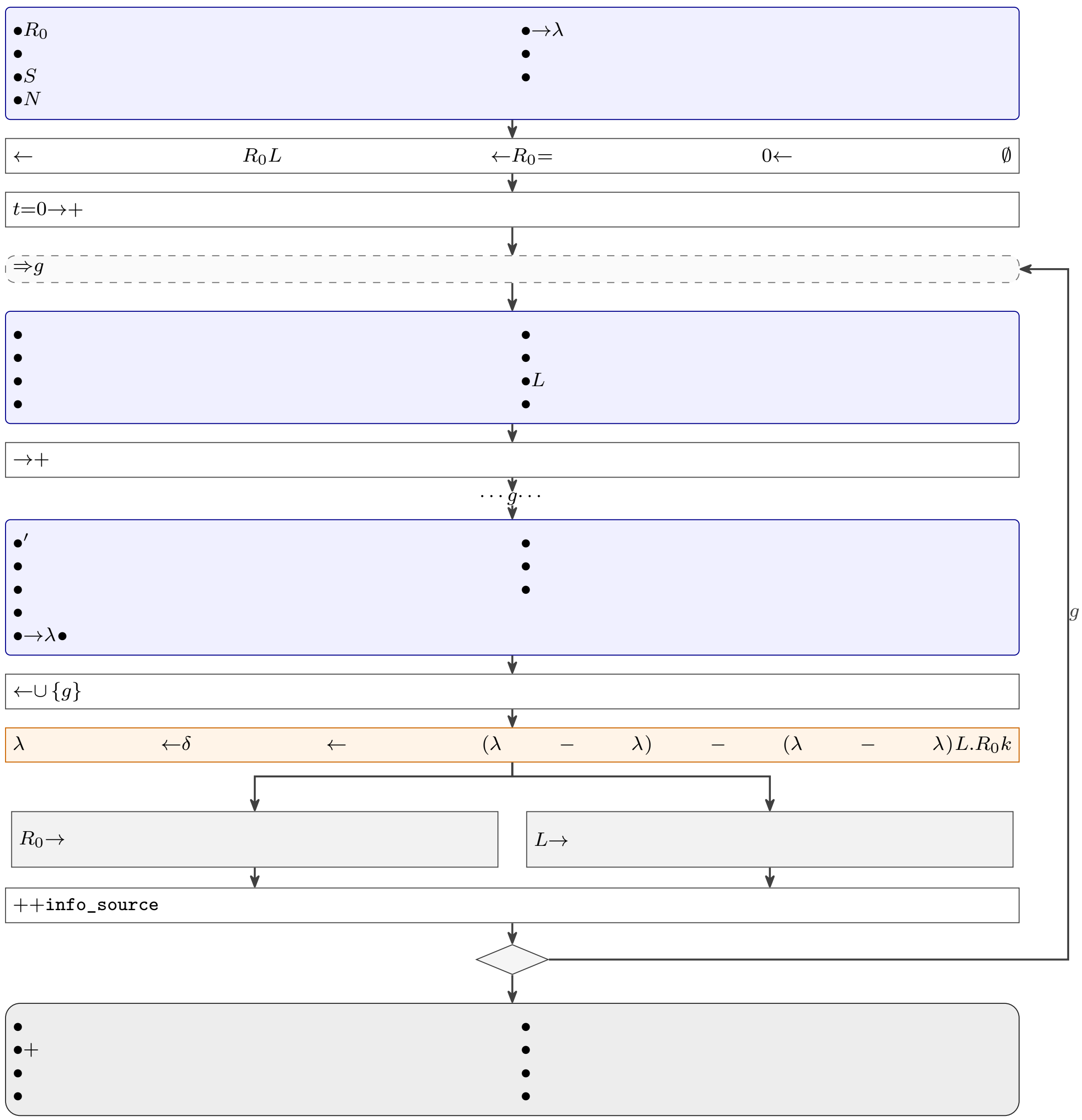


GROUP STAGE
(72 matches, nearly flat)
KNOCKOUTS
the FINAL
0.99 bits
semi-finals
Pivotality: expected forecast swing (bits)
Match (in tournament order)

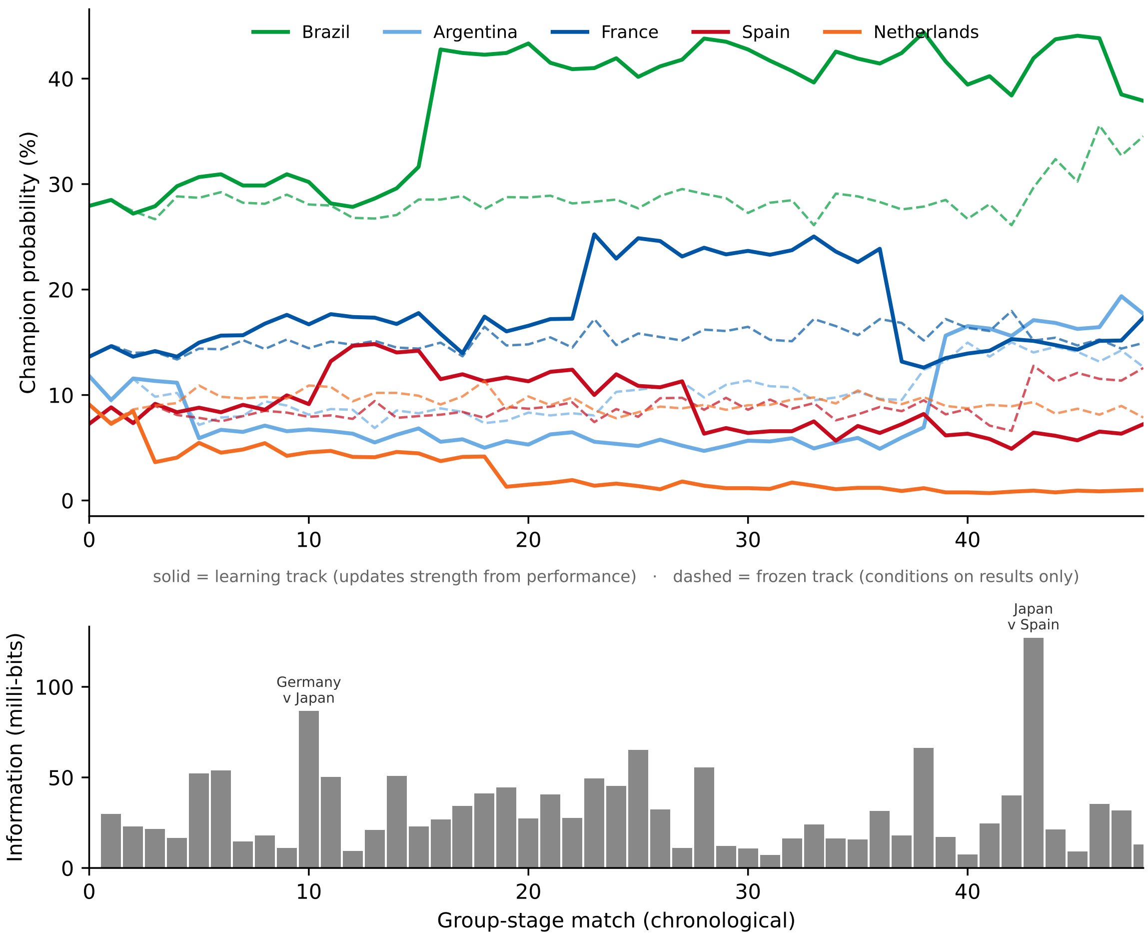
$\{0.05, 0.10, 0.15\}$



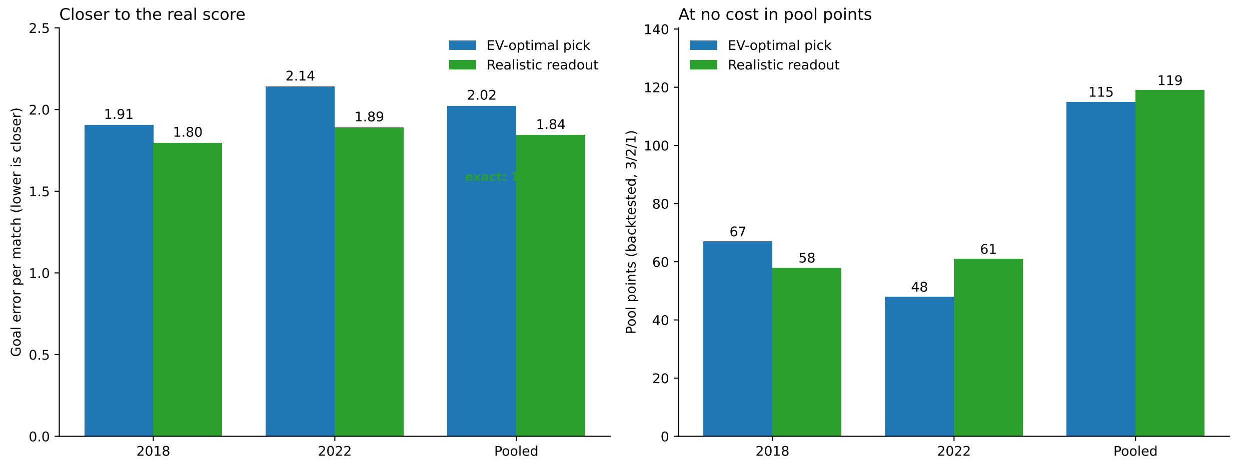
Before the semi-final
If Spain win the semi
If France win the semi
Champion probability, %
0
10
20
30
40
50
60
38
61
0
20
0
51
30
27
34
13
11
15
Spain
France
Argentina
England

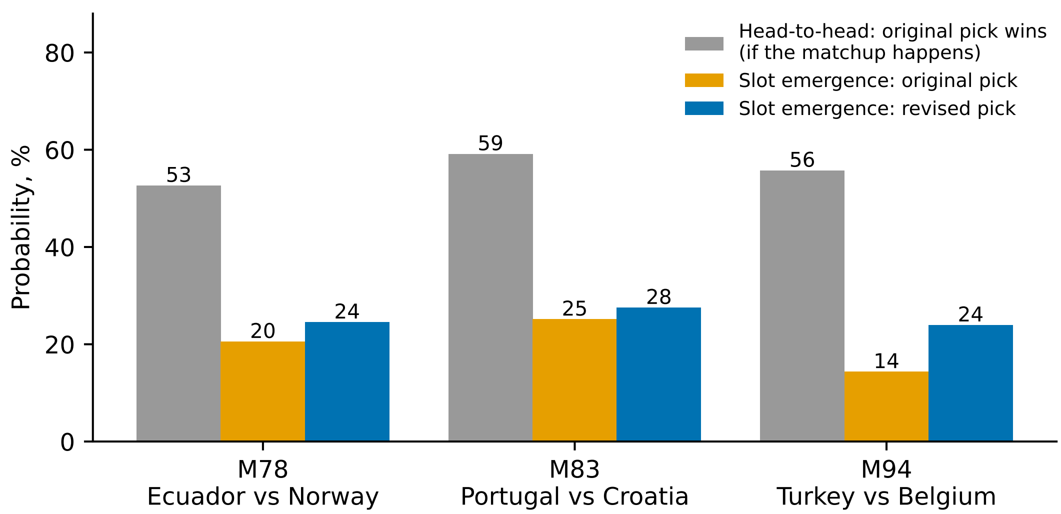


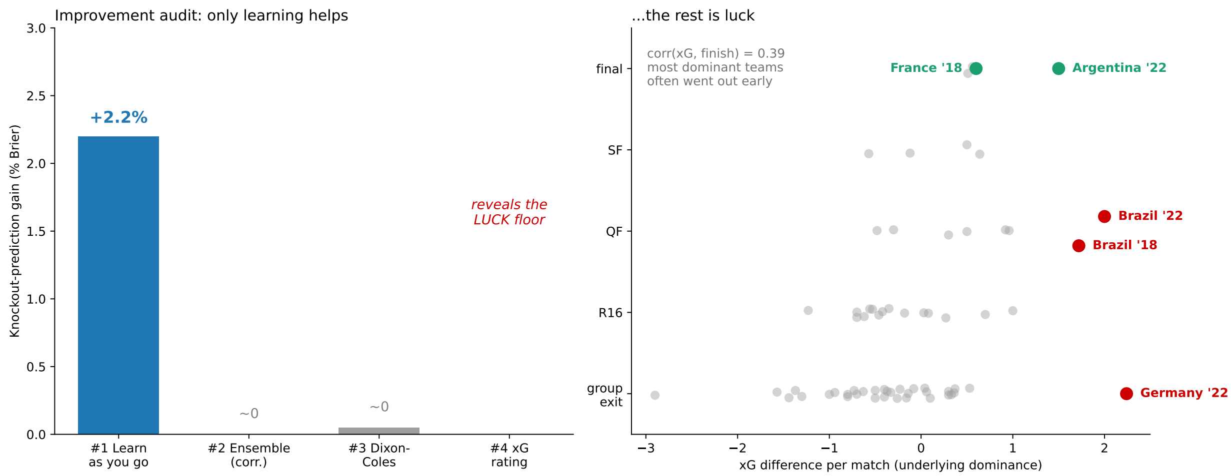
$\lambda\lambda$


Brazil
Argentina
France
Spain
Netherlands
Champion probability (%)
0
10
20
30
40
solid = learning track (updates strength from performance) · dashed = frozen track (conditions on results only)
Information (milli-bits)
0
50
100
Germany v Japan
Japan v Spain
Group-stage match (chronological)

—


Closer to the real score
Goal error per match (lower is closer)
EV-optimal pick
Realistic readout
1.91
1.80
2.14
1.89
2.02
1.84
2018
2022
Pooled
At no cost in pool points
Pool points (backtested, 3/2/1)
EV-optimal pick
Realistic readout
67
58
48
61
115
119
2018
2022
Pooled


Probability, %
Head-to-head: original pick wins
(if the matchup happens)
Slot emergence: original pick
Slot emergence: revised pick
53
20
24
59
25
28
56
14
24
M78
Ecuador vs Norway
M83
Portugal vs Croatia
M94
Turkey vs Belgium

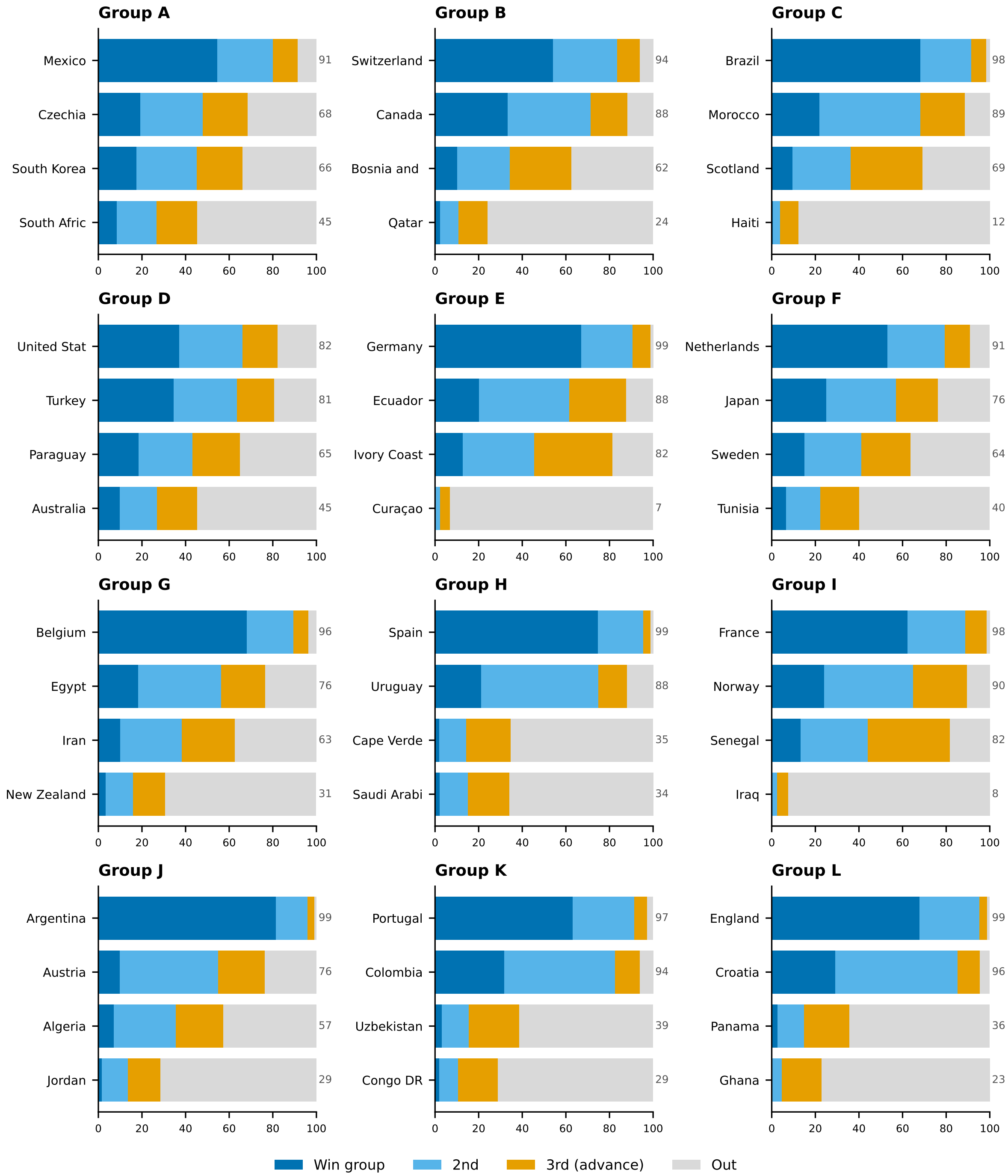


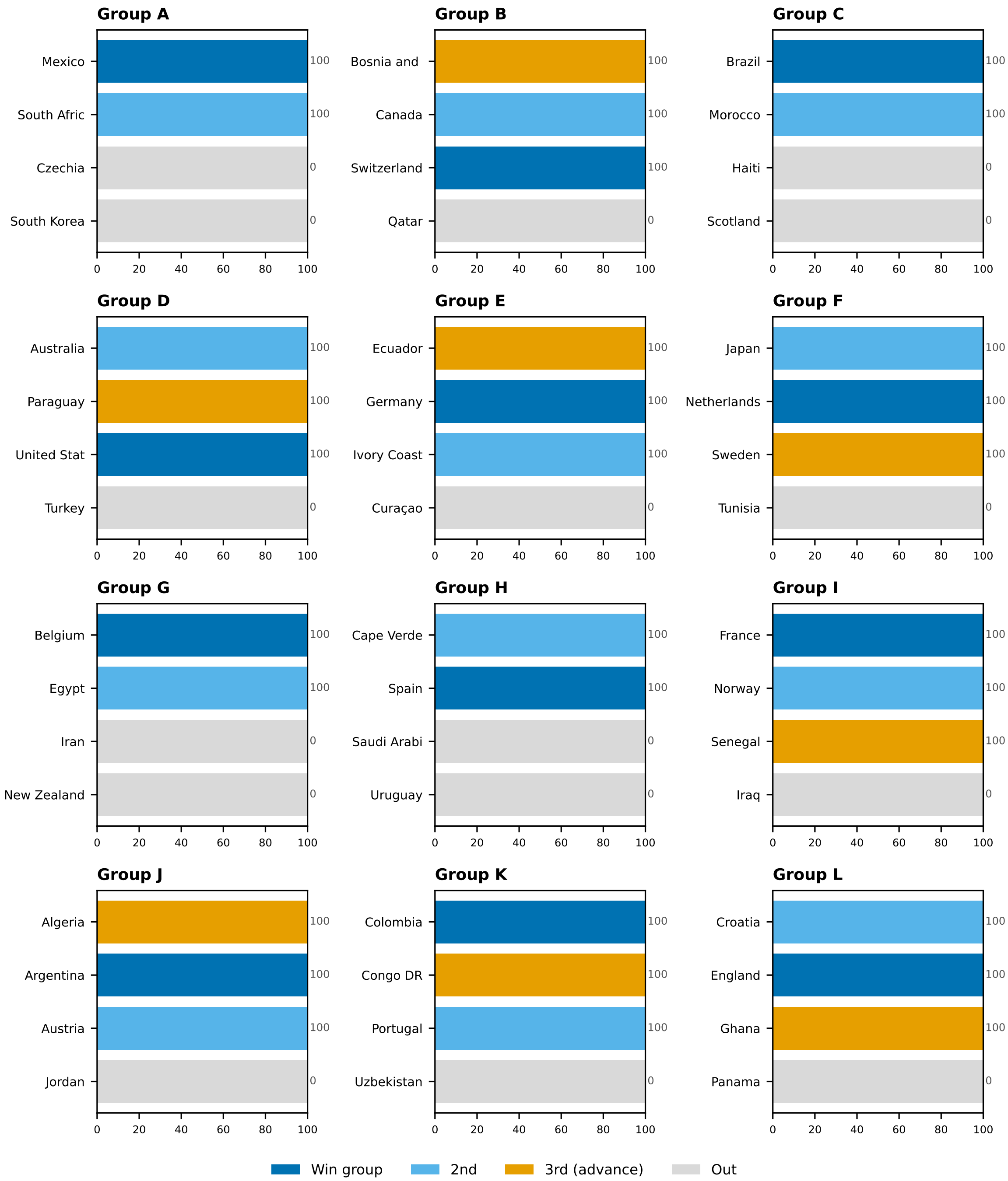
$-20 - 15$

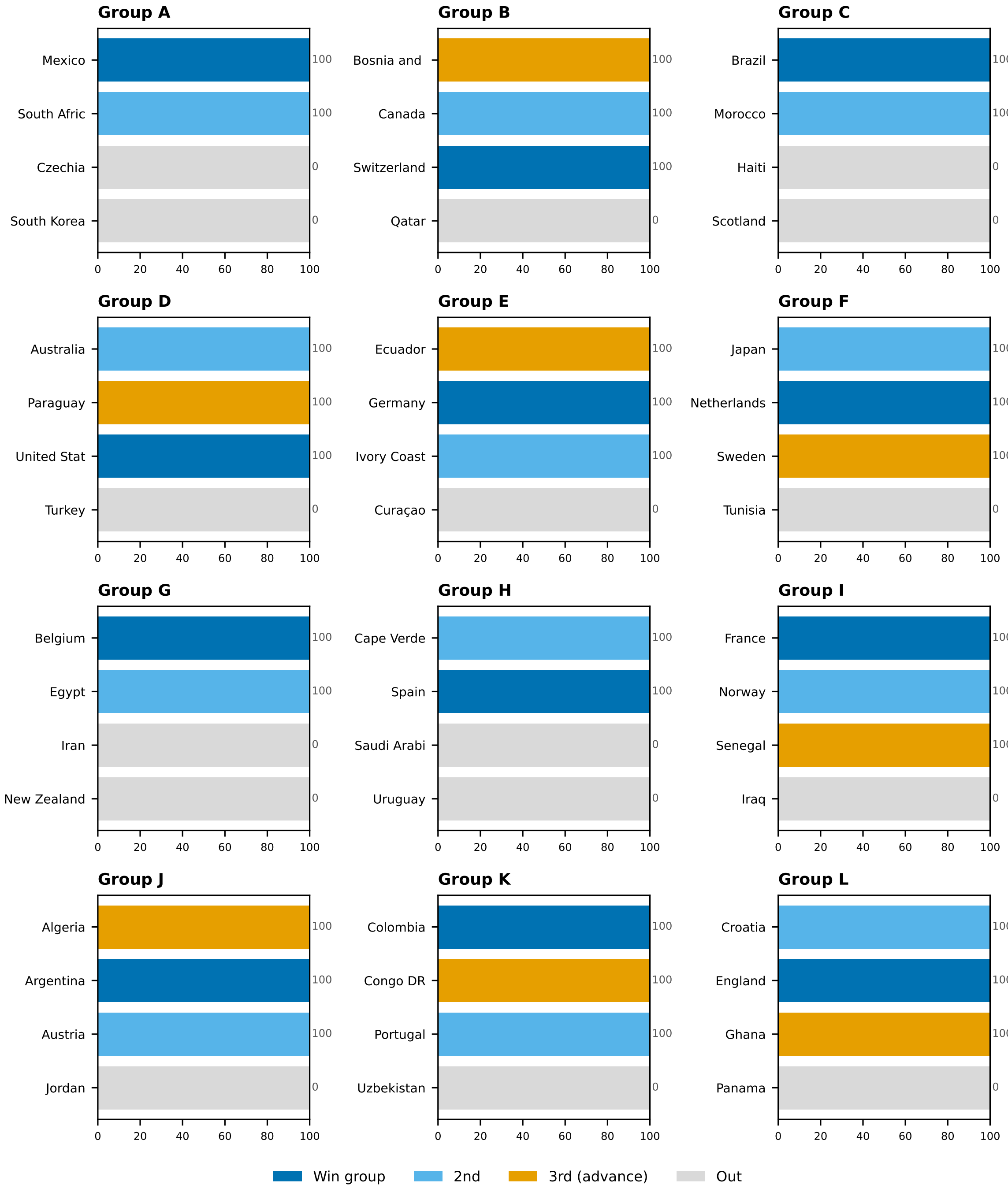
$N = 50{,}000$

$\lambda \in [0.10, 3.50] N = 200{,}000 N = 20{,}000$

$-1 -2 -3$


Group A
Mexico
Czechia
South Korea
South Afric
91
68
66
45
Group B
Switzerland
Canada
Bosnia and
Qatar
94
88
62
24
Group C
Brazil
Morocco
Scotland
Haiti
98
89
69
12
Group D
United Stat
Turkey
Paraguay
Australia
82
81
65
45
Group E
Germany
Ecuador
Ivory Coast
Curaçao
99
88
82
7
Group F
Netherlands
Japan
Sweden
Tunisia
91
76
64
40
Group G
Belgium
Egypt
Iran
New Zealand
96
76
63
31
Group H
Spain
Uruguay
Cape Verde
Saudi Arabi
99
88
35
34
Group I
France
Norway
Senegal
Iraq
98
90
82
8
Group J
Argentina
Austria
Algeria
Jordan
99
76
57
29
Group K
Portugal
Colombia
Uzbekistan
Congo DR
97
94
39
29
Group L
England
Croatia
Panama
Ghana
99
96
36
23
0 20 40 60 80 100
Win group
2nd
3rd (advance)
Out


Group A
Mexico 100
South Afric 100
Czechia 0
South Korea 0
Group B
Bosnia and 100
Canada 100
Switzerland 100
Qatar 0
Group C
Brazil 100
Morocco 100
Haiti 0
Scotland 0
Group D
Australia 100
Paraguay 100
United Stat 100
Turkey 0
Group E
Ecuador 100
Germany 100
Ivory Coast 100
Curaçao 0
Group F
Japan 100
Netherlands 100
Sweden 100
Tunisia 0
Group G
Belgium 100
Egypt 100
Iran 0
New Zealand 0
Group H
Cape Verde 100
Spain 100
Saudi Arabi 0
Uruguay 0
Group I
France 100
Norway 100
Senegal 100
Iraq 0
Group J
Algeria 100
Argentina 100
Austria 100
Jordan 0
Group K
Colombia 100
Congo DR 100
Portugal 100
Uzbekistan 0
Group L
Croatia 100
England 100
Ghana 100
Panama 0
0 20 40 60 80 100
Win group
2nd
3rd (advance)
Out


Group A
Mexico 100
South Afric 100
Czechia 0
South Korea 0
Group B
Bosnia and 100
Canada 100
Switzerland 100
Qatar 0
Group C
Brazil 100
Morocco 100
Haiti 0
Scotland 0
Group D
Australia 100
Paraguay 100
United Stat 100
Turkey 0
Group E
Ecuador 100
Germany 100
Ivory Coast 100
Curaçao 0
Group F
Japan 100
Netherlands 100
Sweden 100
Tunisia 0
Group G
Belgium 100
Egypt 100
Iran 0
New Zealand 0
Group H
Cape Verde 100
Spain 100
Saudi Arabi 0
Uruguay 0
Group I
France 100
Norway 100
Senegal 100
Iraq 0
Group J
Algeria 100
Argentina 100
Austria 100
Jordan 0
Group K
Colombia 100
Congo DR 100
Portugal 100
Uzbekistan 0
Group L
Croatia 100
England 100
Ghana 100
Panama 0
0 20 40 60 80 100
Win group
2nd
3rd (advance)
Out

				
		✓	✓	✓
	✓		×	×
	✓	×		×
	✓	×	×	

✓×

		$\checkmark$	$\checkmark$	$\times$
	$\checkmark$		$\times$	$\checkmark$
	$\checkmark$	$\times$		$\checkmark$
	$\times$	$\checkmark$	$\checkmark$	

$\checkmark\times$
-20

		$\times$	$\checkmark$	$\checkmark$
	$\times$		$\checkmark$	$\checkmark$
	$\checkmark$	$\checkmark$		$\checkmark$
	$\checkmark$	$\checkmark$	$\checkmark$	

$\checkmark\times$

		$\checkmark$	$\checkmark$	$\times$
	$\checkmark$		$\times$	$\times$
	$\checkmark$	$\times$		$\times$
	$\times$	$\times$	$\times$	

$\checkmark\times$

		$\checkmark$	$\times$	$\checkmark$
	$\checkmark$		$\times$	$\checkmark$
	$\times$	$\times$		$\times$
	$\checkmark$	$\checkmark$	$\times$	

$\checkmark\times$
$-15-10$

		$\times$	$\checkmark$	$\checkmark$
	$\times$		$\times$	$\checkmark$
	$\checkmark$	$\times$		$\checkmark$
	$\checkmark$	$\checkmark$	$\checkmark$	

$\checkmark\times$

		$\times$	$\times$	$\checkmark$
	$\times$		$\times$	$\checkmark$
	$\times$	$\times$		$\times$
	$\checkmark$	$\checkmark$	$\times$	

$\checkmark\times$

		$\times$	$\checkmark$	$\checkmark$
	$\times$		$\times$	$\times$
	$\checkmark$	$\times$		$\times$
	$\checkmark$	$\times$	$\times$	

$\checkmark\times$

		✓	✓	✓
	✓		✓	✓
	✓	✓		✓
	✓	✓	✓	

✓×

		✓	✓	✓
	✓		×	✓
	✓	×		✓
	✓	✓	✓	

✓×

		×	✓	✓
	×		×	✓
	✓	×		×
	✓	✓	×	

✓×
-10

		✓	×	✓
	✓		✓	✓
	×	✓		✓
	✓	✓	✓	

✓×

×
✓
✓
×
×
✓
✓
×
✓
✓
✓
✓

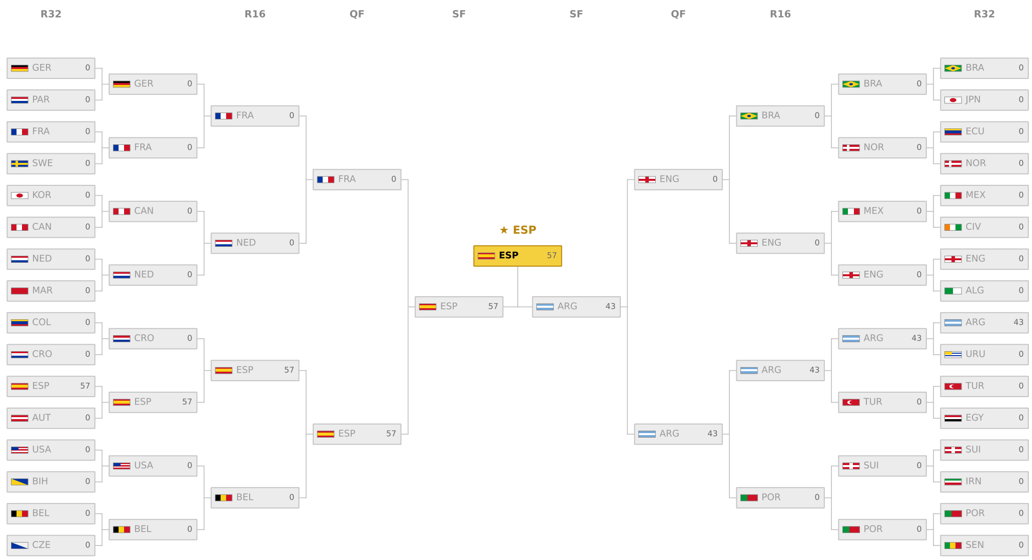
✓×△

* * *

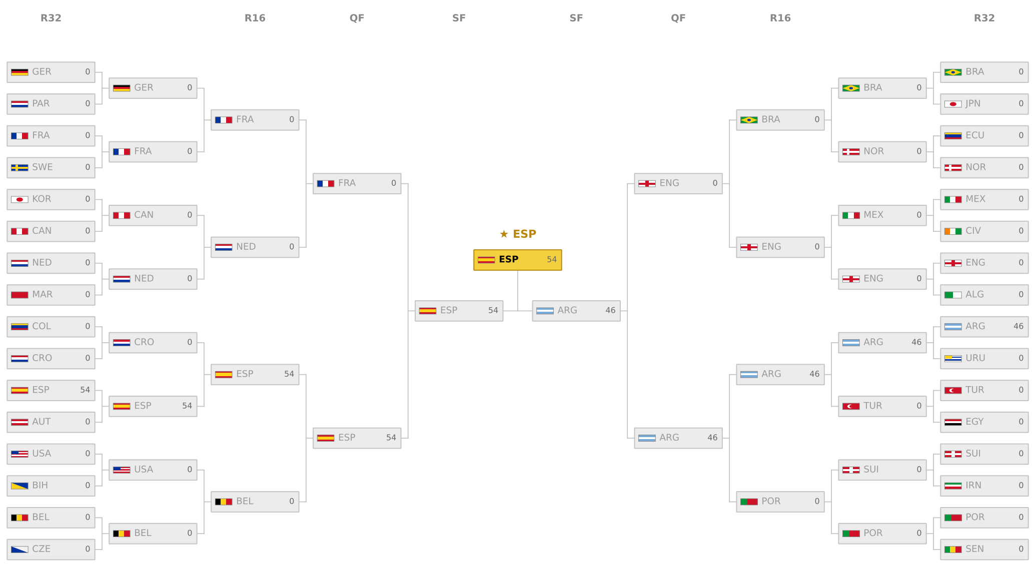
$^*NNN$

011233scripts/make_bracket_tikz.py

Track A: submitted bracket, box value = P(champion) %

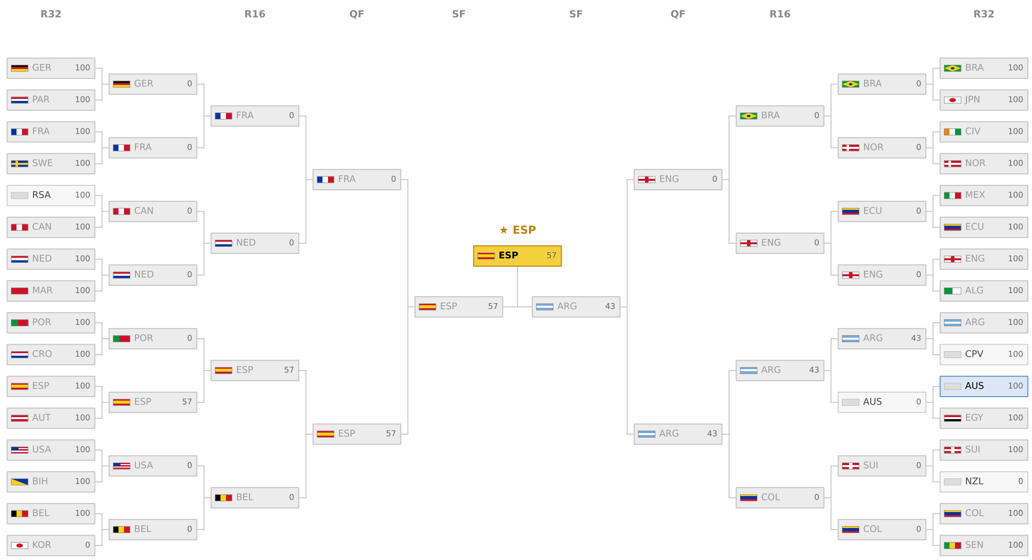


Track B: submitted bracket, box value = P(champion) %

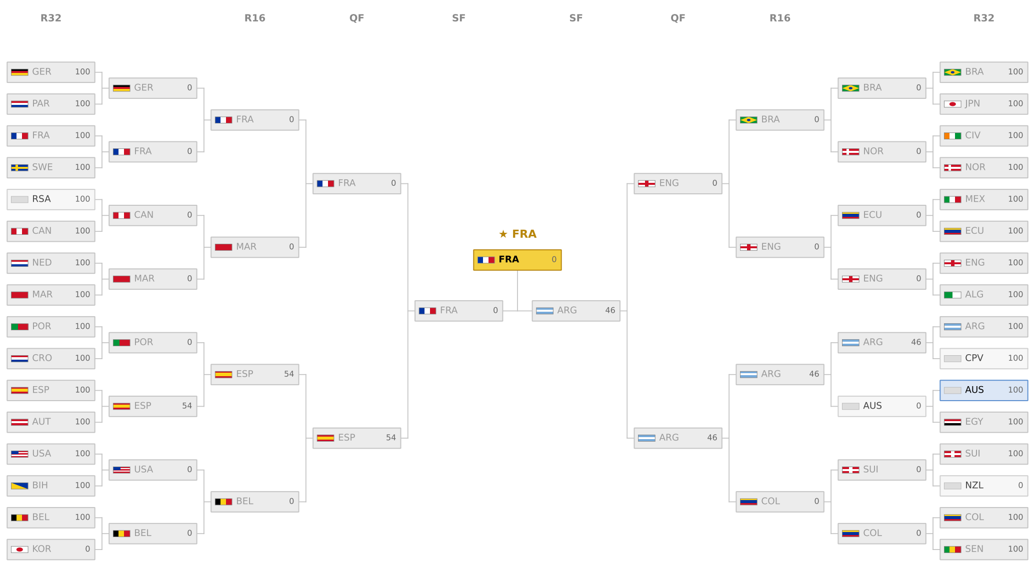


$+4.0{+}2.4$

Track A: most-likely bracket, R32 box = P(reach slot) %, deeper = P(champion) %



Track B: most-likely bracket, R32 box = P(reach slot) %, deeper = P(champion) %



$|\Delta| > 3$

+
+
+
+

k

$kN = 3{,}000 k = 40 k = 0$

k

$N = 3{,}000 k = 60$

$1900 \to 1736$
$1900 \to 1954$
λ
λ
$+75 \pm 75$

$\approx$ $+5.8 +6.6$
$\pm 50 \times 2$ $+4.8 +6.7$
$\lambda \times 0.8 \times 1.2$ $+7.5 +5.3$

λ

$$\lambda \ = \ 0.326 \times (),$$

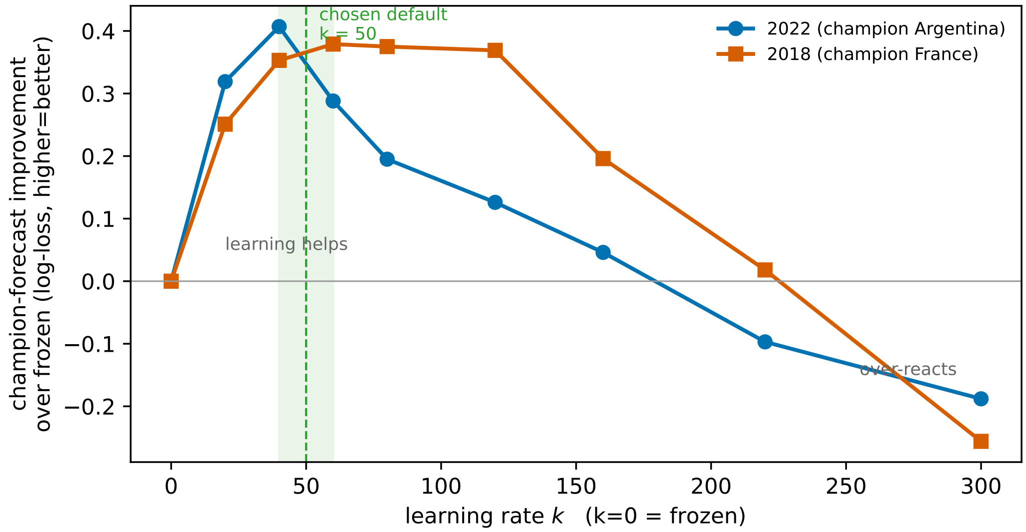
$\Delta e_0 = (1 + 10^{-\Delta/400})^{-1} p_D = 0.30\, e^{-|\Delta|/700} p_H = e_0 - p_D/2 p_A = 1 - p_H - p_D (\lambda, \lambda)$

$$\delta \ = \ \big(\lambda - \lambda\big) \ - \ \big(\lambda - \lambda\big),$$

$$\leftarrow \ \rho \ + \ \big(k\,\delta, \ -b, \ +b\big), \ \ = \ \ + \,,$$

$\rho = 0.95 b = 75 k k = 0 k$

$t = 0$
$kkk = 401.662.077.210.619.937.5 k = 60 k 4060 k = 50 k = 50 \rho = 0.95 b = 75 k$
$kkk = 60 + 7.22.011.76 k = 40 + 1.82.692.41$
$28.5 - 8015.30.075023.00.38.918.520892161$
$\pm 50 \pm 204.87.5$
$+6.8 - 5.3 \pm 6.7 + 2.5 + 0.6 \pm 4.70.3260.3140.991.091.181.32$
collect_matchperformancefetch_xgcalibrate_proxyλlearnsim_tournamentsnapshotreplay$k$
tournament_2018/2022run_2022_replayrun_2018_ksweepsweep_kpilotstress_test$k = 500.95$
75



$kk4060kk = 50$

https://doi.org/10.1086/677350

https://doi.org/10.1198/016214506000001437

https://doi.org/10.1109/TSSC.1966.300074

https://doi.org/10.1111/rssa.12042

https://doi.org/10.1515/1559-0410.1418

https://doi.org/10.1007/s10994-018-5703-7

https://doi.org/10.1287/opre.1070.0448

https://doi.org/10.1111/1467-9876.00065

https://doi.org/10.1515/jqas-2018-0060

https://doi.org/10.1515/jqas-2014-0051

https://doi.org/10.1016/j.ejor.2014.10.054

https://doi.org/10.1111/1467-9884.00243

https://doi.org/10.1080/00031305.1997.10473945

https://doi.org/10.1016/j.ijforecast.2009.10.002

https://doi.org/10.1287/mnsc.47.3.369.9769

https://doi.org/10.1111/1467-9884.00366

https://doi.org/10.1016/j.ijforecast.2009.10.001

https://doi.org/10.1111/j.1467-9574.1982.tb00782.x

https://doi.org/10.1016/j.ijforecast.2014.02.008